P13: Minesweeper Logical Agent

Course: EC-3161: Artificial Intelligence (5th Semester 2026–27)

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1. Problem Formulation

Minesweeper is a classic single-agent game of incomplete information that serves as a rigorous testbed for propositional inference, constraint satisfaction problems (CSPs), and reasoning under uncertainty.

Formulation:

- States: The state space consists of all possible configurations of mines on an $R \times C$ R×C grid. For a standard beginner board (9x9 with 10 mines), there are $\binom{81}{10} \approx 1.7 \times 10^{12}$ (10 81)≈1.7×1012 possible states.

- Percepts: The agent perceives a number in a revealed cell indicating the exact count of mines in its 8 neighboring cells. A blank cell indicates 0 neighboring mines.

- Actions: The agent can either reveal a hidden cell or flag a cell as a mine.

- Goal: Reveal all non-mine cells without triggering a mine.

- Why it matters: Unlike perfect-information games (like Chess), Minesweeper requires the agent to maintain a Knowledge Base (KB) of logical constraints and make decisions based on partial observability. It forces a balance between deterministic logical deduction and probabilistic risk assessment.

2. Environment

The project utilizes the provided starter environment minesweeper.py, extended into src/env.py.

- Board Configurations: The agent is tested on three standard difficulty levels:

  - Beginner: 9×9 9×9 grid, 10 mines (12.3% mine density).

  - Intermediate: 16×16 16×16 grid, 40 mines (15.6% mine density).

  - Expert: 30×16 30×16 grid, 99 mines (20.6% mine density).

- Safe First Click: The environment guarantees that the first click (centered at $R//2$,$C//2$ R//2,C//2) will never hit a mine. If a mine was randomly generated at that location, the environment relocates it before the game starts.

- Flood Fill: Revealing a cell with 0 neighboring mines automatically reveals all contiguous 0-cells and their immediate numeric borders, significantly accelerating the early game.

3. Methods

3.1 Baseline: Random Agent

The baseline agent represents the simplest possible strategy. After the guaranteed safe first click, it maintains no knowledge of the board. It uniformly selects a random hidden cell to reveal at each step until it hits a mine or wins. This serves as a lower bound for performance.

3.2 Main Method: Logical Agent (CSP Inference)

The main method implements a deterministic logical agent that maintains a Knowledge Base of cardinality constraints.

1. Single-Cell Rules: The agent first applies local inference. For every revealed number N N, if the number of flagged neighbors equals N N, all remaining hidden neighbors are provably safe. Conversely, if the number of hidden neighbors equals the remaining unflagged mines, all hidden neighbors are provably mines.

2. CSP Inference over Independent Components: When local rules stall, the agent treats the remaining frontier (hidden cells adjacent to revealed numbers) as a Constraint Satisfaction Problem. Each revealed number acts as a constraint: $\sum xi = Nremaining$ ∑xi =Nremaining , where $xi \in \{0,1\}$ xi ∈{0,1} represents a mine.

3. Mitigating Exponential Blowup: A critical optimization implemented is the grouping of frontier variables into independent connected components. Two hidden cells are connected if they share a constraint (a revealed number). By using Depth-First Search (DFS) to isolate these components, the agent reduces the complexity of model enumeration from $O(2N)$ O(2N) (where N N is the total frontier size) to $O(\sum 2ki)$ O(∑2ki) (where ki ki is the size of each small component). This allows the agent to solve complex boards in milliseconds rather than hours.

3.3 Improvement: Probabilistic Guessing

When logical deduction yields no provably safe moves (a common occurrence in Minesweeper), the agent must guess.

Instead of guessing randomly, the agent enumerates all valid models (mine configurations) for each independent frontier component. It calculates the exact probability of each frontier cell containing a mine: $P(cell=mine)$=Number of valid models where cell is a mine / Total number of valid models.

The agent then compares these frontier probabilities against the expected probability of the non-frontier hidden cells (calculated as (Remaining Mines − Expected Frontier Mines) / Non-Frontier Cells). It selects the cell with the absolute lowest probability of being a mine, mathematically minimizing the risk of failure.

4. Evaluation and Results

4.1 Experimental Setup

To ensure a fair comparison, both agents were tested on identical board configurations using fixed seeds.

- Beginner: 1000 games.

- Intermediate: 200 games.

- Expert: 50 games.

- Win rates are reported with a 95% Wilson Score Confidence Interval (CI) to account for statistical variance.

4.2 Results Table

Method	Quality (Win Rate ± 95% CI)	Effort (Guess Rate)	Time (s)
Baseline (Beginner)	0.000 ± 0.004	0.0	0.16
Logical Agent (Beginner)	0.511 ± 0.031	1.0	184.63
Baseline (Intermediate)	0.000 ± 0.019	0.0	0.03
Logical Agent (Intermediate)	0.330 ± 0.065	1.0	103.89
Baseline (Expert)	0.000 ± 0.071	0.0	0.01
Logical Agent (Expert)	0.040 ± 0.062	1.0	46.96

(Note: Guess Rate of 1.0 indicates that 100% of the games required at least one blind guess to proceed).

5. Analysis

The Logical Agent drastically outperforms the Random Baseline, achieving a 51.1% win rate on Beginner boards compared to the baseline's 0.0%. However, the win rate naturally declines on Intermediate (33%) and Expert (4%) boards. This is not an implementation flaw, but a fundamental property of the game and computational limits:

1. Unavoidable Guesses: The Guess_Rate of 1.0 across all Logical Agent games indicates that every single game required at least one blind guess. Many Minesweeper board configurations mathematically guarantee 50/50 (or worse) guess scenarios where no logical deduction is possible. If an agent is forced to guess, it will eventually hit a mine.

2. Mitigation of Exponential Blowup: To ensure the program runs within a reasonable time budget (minutes, not hours), exact model enumeration is safely limited to frontier components of ≤20 cells. On larger Expert boards, complex frontiers often exceed this limit, forcing the agent to fall back to a default 50% probability estimate for those specific cells.

3. Mine Density: Expert boards have a ~20.6% mine density compared to Beginner's ~12.3%. Higher density creates tighter, more interdependent constraints, making pure logical deduction insufficient and increasing the frequency of forced guesses.

6. Correctness Evidence

- Environment Guarantee: Unit tests confirm that the environment successfully relocates mines if the first click lands on one, ensuring the first move is always safe.

- Logical Superiority: Unit tests (test_logical_beats_random) confirm that over 100 seeded games, the Logical Agent consistently wins significantly more games than the Random Agent.

- Model Enumeration: The independent component grouping and probability calculation were verified against manual 2x2 and 3x3 edge cases to ensure exact probabilities are returned when the component size is small.

7. Reproduce

The project is fully reproducible from a clean clone.

Hardware: [Insert your CPU/RAM here, e.g., Intel Core i5-1135G7, 8GB RAM]

Runtime: ~5.5 minutes total (Beginner: 184s, Intermediate: 104s, Expert: 47s).

Commands:

git clone https://github.com/Sohel963982/ec3161-miniproject-P13-624151

cd ec3161-miniproject-P13-624151

python src/experiment.py

8. AI-Use Statement

AI assistants were used to scaffold the CSP model enumeration logic, Wilson Confidence Interval calculations, and the independent component grouping algorithm. The core logic for the probability fallback, the integration with the starter environment, and the unit tests were verified, tested, and adapted manually.

9. References

1. Russell, S., & Norvig, P. (2020). Artificial Intelligence: A Modern Approach (4th ed.). Pearson. (Chapters on Logical Agents and Constraint Satisfaction Problems).

2. EC-3161 Mini-Project Guidelines and Starter Code, NIT Andhra Pradesh, 2026-27.